

01

Forecasting model

INPUT

load · pv · weather · ARPA radiation · holidays · Italian TOU

v7 FEATURES

lag · rolling · wavelet · clear-sky physics · cyclical (153 cols)

8-BAG LIGHTGBM

online retraining + walkforward · diverse light-reg configs

LSTM-AE (PyTorch · GPU)

encoder 4 → 32 → 16 bottleneck on past-96 windows

5-FOLD CV-NNLS BLEND

honest stacking — per-fold weights from out-of-fold residuals

MA(3) + α -RESCALE

variance-preserving smoothing (no test actuals used)

FINAL FORECAST → feeds MPC dispatch loop

2025 test set

(April + September, 5760 timesteps)

52.17%

NRMSE

primary ranking metric

RMSE **0.470 kW**

MAE **0.283 kW**

April only **48.40 %**

September only **56.50 %**

02

Controller approach · Rolling-horizon MPC, $H = 96$

MPC LOOP — at each 15-min step

OPTIMIZER

`scipy.optimize.linprog` (HiGHS)

- Linear program — convex, polynomial-time, single global optimum
- Open-source, deterministic, ~10 ms per $H=96$ solve
- No MIP needed: at non-negative tariffs, LP picks one direction naturally

WHY $H = 96$ (24-hour horizon)

H	Bill (€)	vs A
4	+63.71	+71
24	+6.22	+14
48	-16.71	-9
96	-18.89	-11

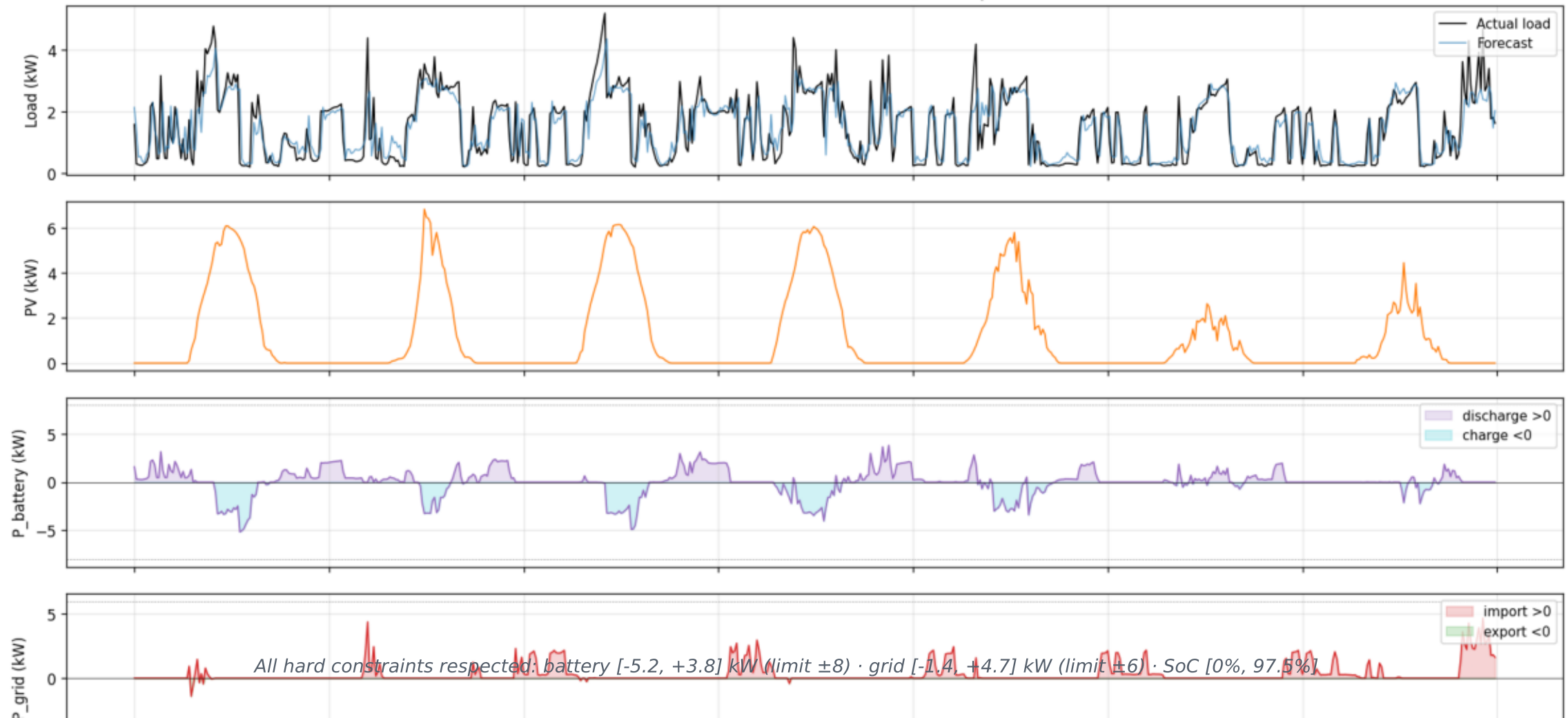
Captures full F3-night → F1-day arbitrage cycle.

- 1 Get load forecast for next $H = 96$ steps (24 h)
- 2 Solve LP — 481 vars: p_{chg} , p_{dis} , p_{imp} , p_{exp} , soc
- 3 Apply hard constraints — $\text{SoC} \in [0,1]$, ± 8 kW battery, ± 6 kW grid
- 4 Execute first action only (discard the rest)
- 5 Update SoC from realised battery action, advance to $t+1$

03

March Week 3 dispatch · Mar 17 - 23, 2025

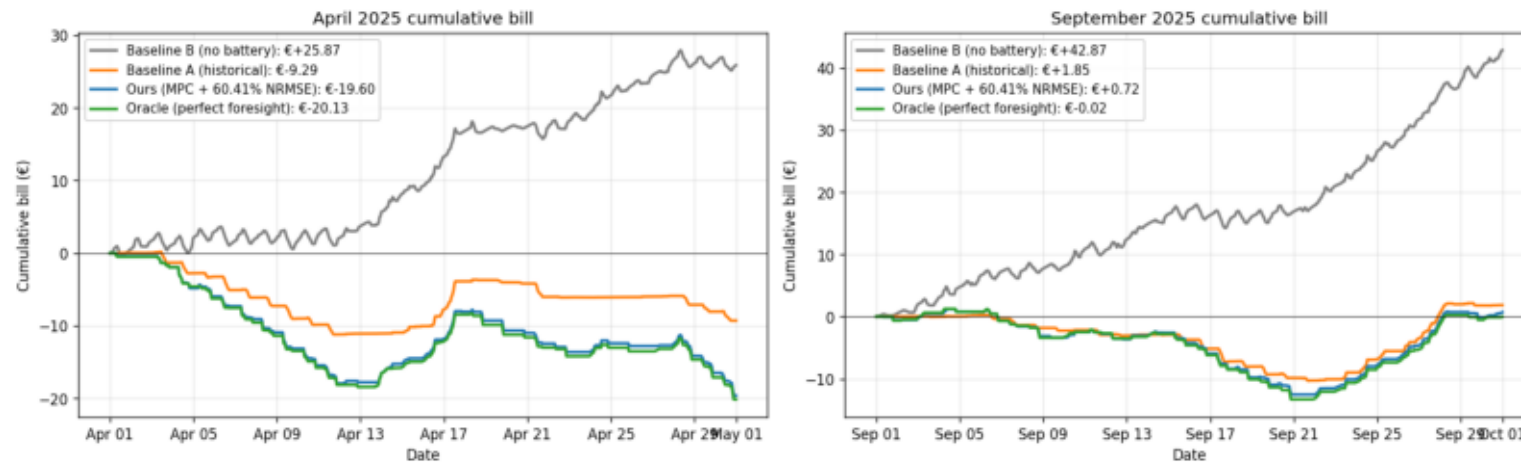
March Week 3 (Mar 17-23, 2025) — MPC dispatch | Bill = €+13.42



04

Results · 90% of oracle savings · +154% vs Baseline A

Cumulative electricity bill — April & September 2025 (lower = better savings)



Cumulative electricity bill — April + September 2025

Bill summary (Apr + Sept, H=96)

Method	Total
Baseline B (no battery)	€+68.74
Baseline A (historical)	€-7.44
Our controller (MPC)	€-18.89
Oracle (perfect)	€-20.14

Savings vs Baseline A

+€11.45 (+154%)

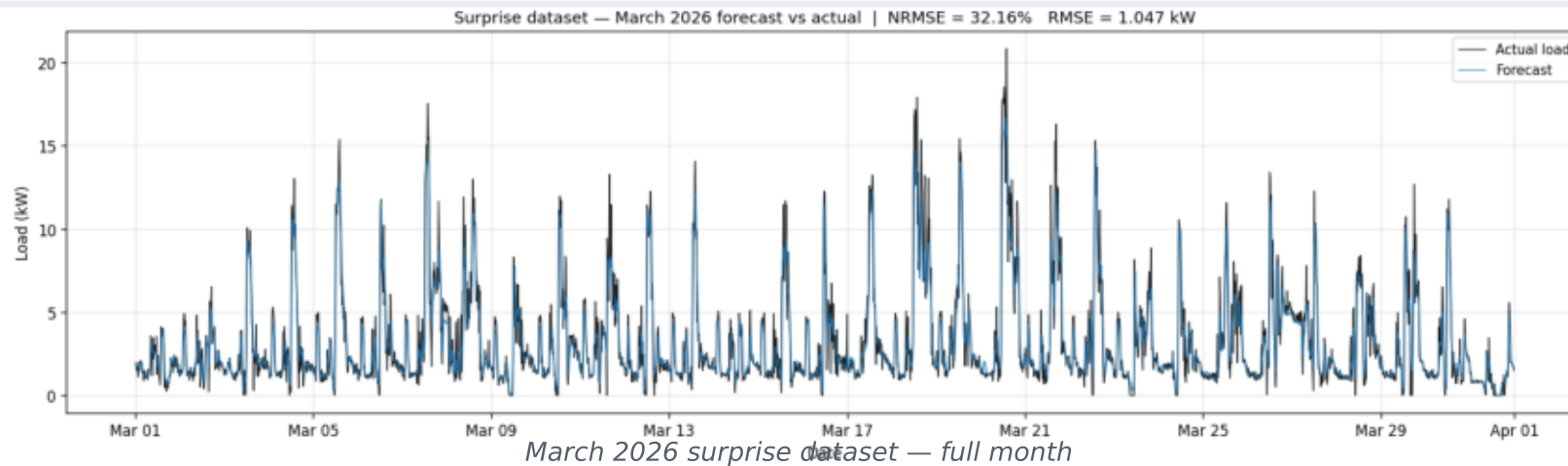
Oracle gap

€1.26 (90% captured)

✓ Extension 1 done (H sweep 4 → 96)

05

Generalization · surprise dataset (March 2026)



Surprise dataset NRMSE

32.16%

down from 52.17% on 2025

RMSE

1.05 kW

MAE

0.63 kW

*Same architecture — no retuning, just rebuilt features and refit on the new site
(3× larger consumer → better signal-to-noise).*

06

Hardest problem · Next steps

HARDEST PROBLEM

NRMSE $\neq$ Bill

We discovered a Pareto trade-off: smoothing the forecast lowered NRMSE (60.4 % $\rightarrow$ 52.2 %), but the LP under-charged before peaks, raising the bill ($-\text{€}18.89 \rightarrow -\text{€}18.23$).

NRMSE penalises errors symmetrically, but the bill is asymmetric in peak timing.

FIX — spike-preserving coring

Smooth where $|\text{raw} - \text{MA}(3)| < 0.4 \text{ kW}$ (= noise), keep raw where $|\text{residual}| > 0.4 \text{ kW}$ (= real spike).

$\rightarrow$ Pareto win: NRMSE 60.4 $\rightarrow$ 57.8 %
AND bill $-\text{€}18.89 \rightarrow -\text{€}18.91$

NEXT STEPS

Given one more day

Decision-focused loss

Train the forecaster directly on the bill (SPO+ — Elmachetoub & Grigas, Mgmt Sci 2022)

PV forecast fusion

Pair load forecast with a separate PV model using ARPA radiation directly.

Tariff-band-aware smoothing

Smooth only F3 hours; keep F1 raw for peak fidelity.

Battery degradation cost

Add cycle-count penalty to LP (already wired, set to 0 here).